ASR Model Selection Report

Egyptian Arabic speech recognition — bake-off results

Best robot realtime	Best accuracy	Best speed	Models OK
Whisper-Large-v3-Turbo-CT2	**Whisper-Large-v3-CT2**	**MMS-1B-all**	**12**

Overview

12 ASR models were evaluated successfully on the Arabic test clip.
Robot realtime score weights: accuracy 45% + speed/RTF 30% + low VRAM 15% + fast load 10%.
RTF (Real-Time Factor) = transcribe_time / audio_duration. LOWER better. RTF < 1 = realtime.
WER (Word Error Rate): LOWER better. Accuracy %: HIGHER better (approx. 1 - WER).
Top composite pick: Whisper-Large-v3-Turbo-CT2.
Top accuracy pick: Whisper-Large-v3-CT2.

How to read this report

Page 2: metrics glossary — RTF, WER, CER, VRAM, and what to prefer.
Page 3: recommended picks with rationale.
Pages 4-6: accuracy, speed, WER, and accuracy-RTF tradeoff charts.
Pages 7-8: VRAM/load and normalized score breakdown.
Page 9: full accuracy ranking and selection guidance.

ASR - Metrics glossary (what to prefer)

Use this page while reading the charts. Direction labels show what you want for a production robot.

Accuracy % (word accuracy)

HIGHER better

Share of words recognized correctly (about 1 - WER). Higher = fewer recognition mistakes.

WER - Word Error Rate

LOWER better

Fraction of words wrong. 0 = perfect. Lower = better recognition.

CER - Character Error Rate

LOWER better

Same idea as WER at character level. Useful for Arabic. Lower = better.

Transcribe time (s)

LOWER better

Wall time to finish transcription. Lower = faster handoff to the LLM.

Realtime capable

Prefer YES

Yes when RTF < 1. Needed for near-live listening on the robot.

Peak VRAM (MB)

LOWER better

GPU memory used at peak. Lower leaves room to run ASR + LLM + TTS together on one GPU.

Peak RAM (MB)

LOWER better

System (CPU) memory used at peak. Lower is safer on smaller servers / VPS hosts.

Peak CPU %

LOWER better

CPU utilization spike during the run. Lower means less contention with other services.

Cold load time (s)

LOWER better

Seconds to load the model into memory the first time. Matters for startup / cold starts.

RTF (Real-Time Factor)

LOWER better

Processing time / audio duration. RTF < 1 = faster than realtime (good). RTF > 1 = slower than the audio.

x Realtime (xRT)

HIGHER better

How many times faster than realtime (about 1/RTF). Example: 6x = 6s of audio in ~1s of compute.

Robot realtime score (0-100)

HIGHER better

Composite score for robot use (speed + resources + accuracy where relevant). Higher = better fit.

Balanced score (0-100)

HIGHER better

Normalized blend of the main tradeoffs. Higher = better all-rounder.

Success rate %

HIGHER better

Share of runs that completed without error. Prefer 100% for production reliability.

Remember

For ASR: want HIGHER accuracy %, HIGHER robot score, HIGHER xRealtime;
want LOWER WER, LOWER CER, LOWER RTF, LOWER VRAM/RAM, LOWER load/transcribe time.
Prefer realtime-capable (RTF < 1) models for live robot listening.

Recommended picks (automated)

Best for robot realtime (HIGHER score better)

Whisper-Large-v3-Turbo-CT2

Highest composite robot score (accuracy + speed + VRAM + load).

score_robot_realtime=89.62 | avg_accuracy_percent=68.4 | avg_rtf=0.046 | peak_vram_mb=2492.2

Best accuracy % (HIGHER better)

Whisper-Large-v3-CT2

Highest average word accuracy / lowest WER.

avg_accuracy_percent=72.92 | avg_wer=0.2708 | avg_cer=0.1132

Best speed / lowest RTF (LOWER better)

MMS-1B-all

Lowest average RTF (fastest relative to audio length).

avg_rtf=0.044 | avg_x_realtime=22.88 | avg_transcribe_seconds=5.64

Lowest peak VRAM (LOWER better)

Whisper-Small-CT2

Lowest peak VRAM — useful for 6–16 GB GPUs or multi-model co-residency.

peak_vram_mb=1340.2

Best balanced

Whisper-Large-v3-Turbo-CT2

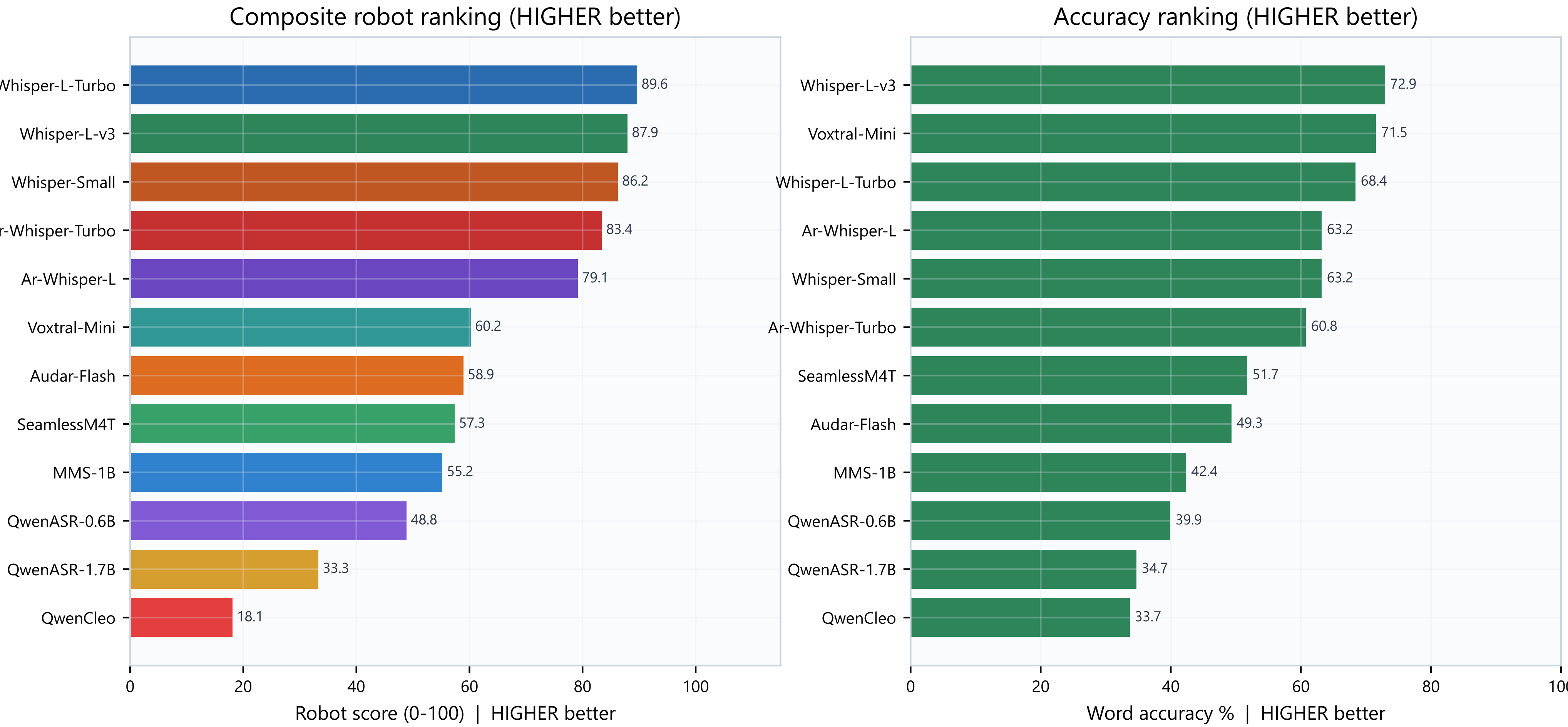
Balanced accuracy/speed/VRAM tradeoff.

score_balanced=90.17

Selection notes

Prefer LOW RTF + HIGH accuracy + LOW VRAM so LLM/TTS still fit on the same GPU.
Whisper CT2 (faster-whisper) models dominate the top of the robot leaderboard.
Arabic fine-tunes were competitive on speed but did not beat base Whisper Large on accuracy here.
Voxtral-Mini has strong accuracy but HIGH VRAM — costly for co-residency.

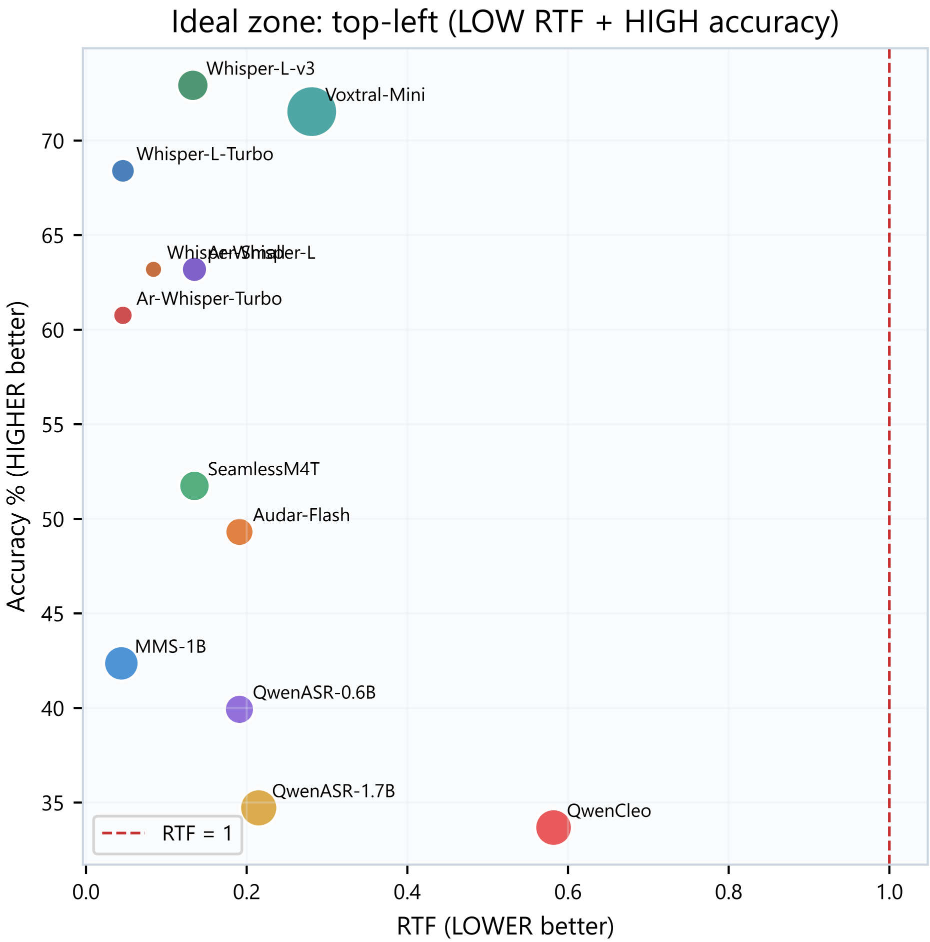
Robot score & Accuracy

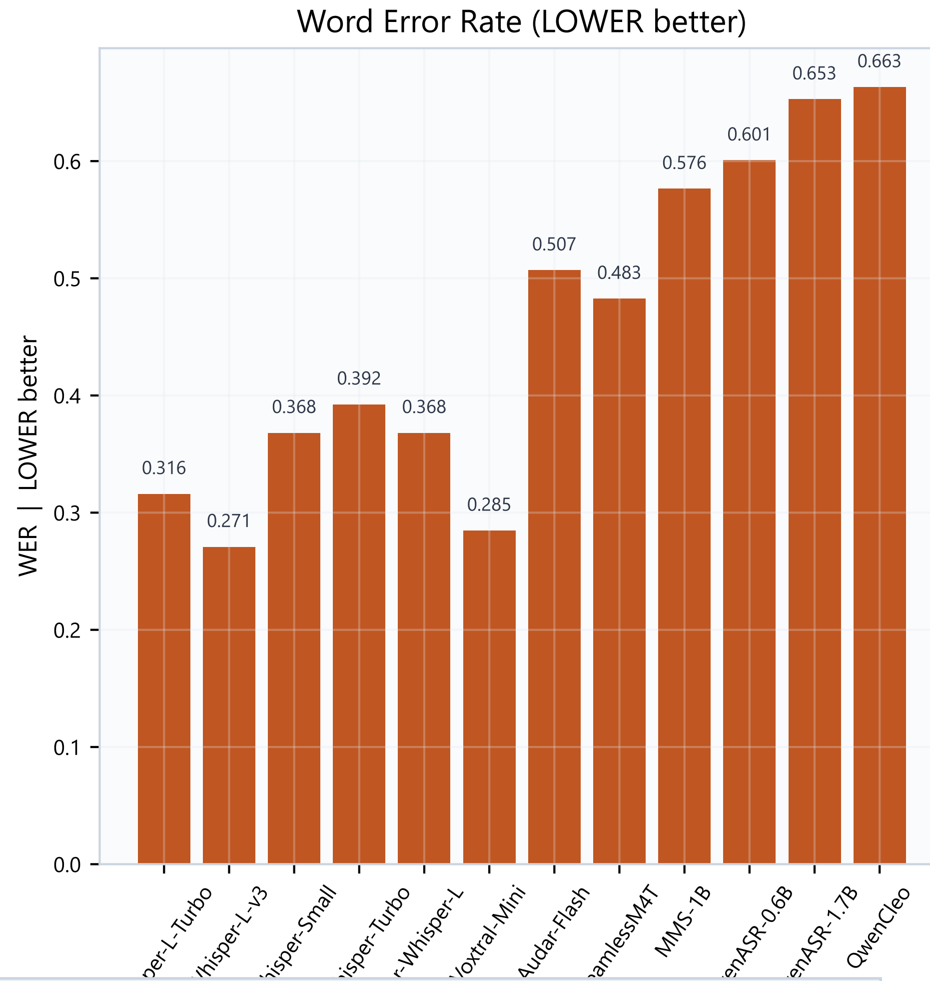


What this means

Left: overall robot fitness (HIGHER better). Right: pure recognition accuracy (HIGHER better).
Whisper-Large-v3-Turbo leads robot score by combining solid accuracy with excellent RTF and moderate VRAM.
Whisper-Large-v3 leads accuracy (72.92%).
Smaller Whisper variants trade some accuracy for LOWER VRAM — useful on tight GPUs.
Bottom of the list (Qwen ASR family / QwenCleo) is too weak on accuracy for production Egyptian Arabic.

Accuracy–speed tradeoff & WER

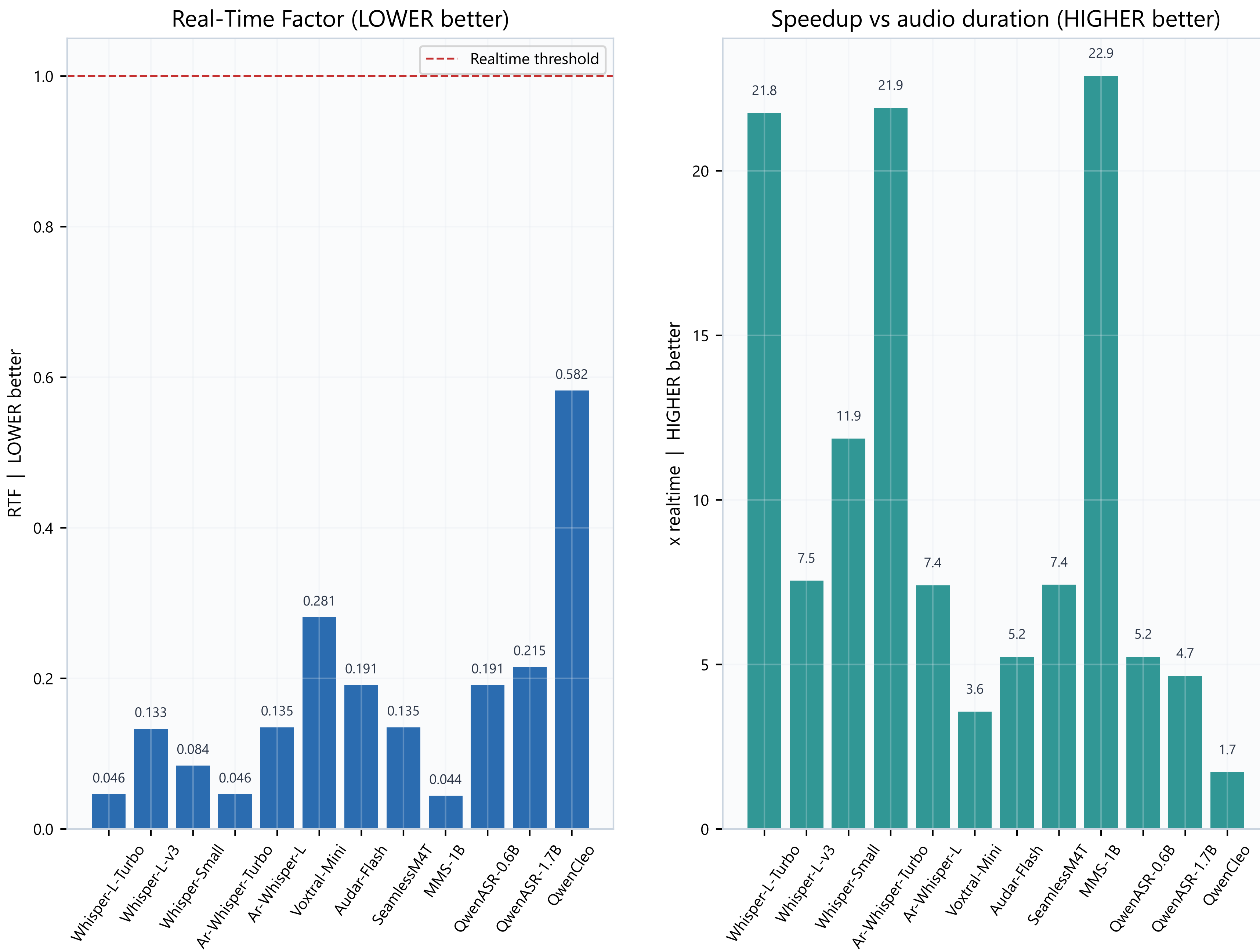




Tradeoff reading

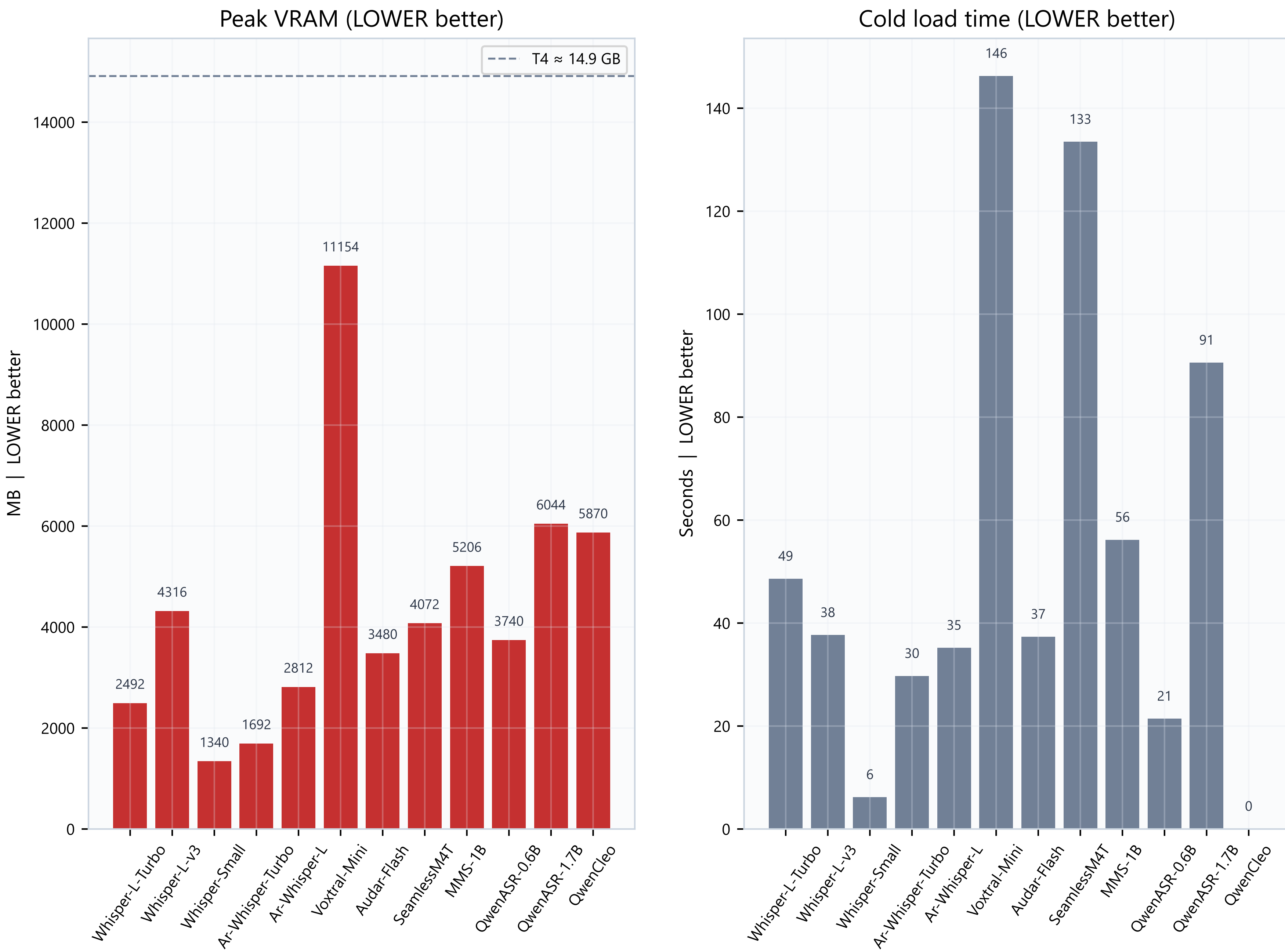
RTF = transcribe_time / audio_length (LOWER better). RTF < 1 = realtime.
WER = Word Error Rate (LOWER better). Accuracy % ~ (1 - WER) x 100 (HIGHER better).
Bubble size = peak VRAM (LOWER better for co-residency with LLM/TTS).
Top-left models are both accurate and fast — preferred for voice robots.
All evaluated models were realtime-capable (RTF < 1) on this clip.
Voxtral is accurate but HIGH VRAM — expensive to host beside an LLM.

Speed metrics



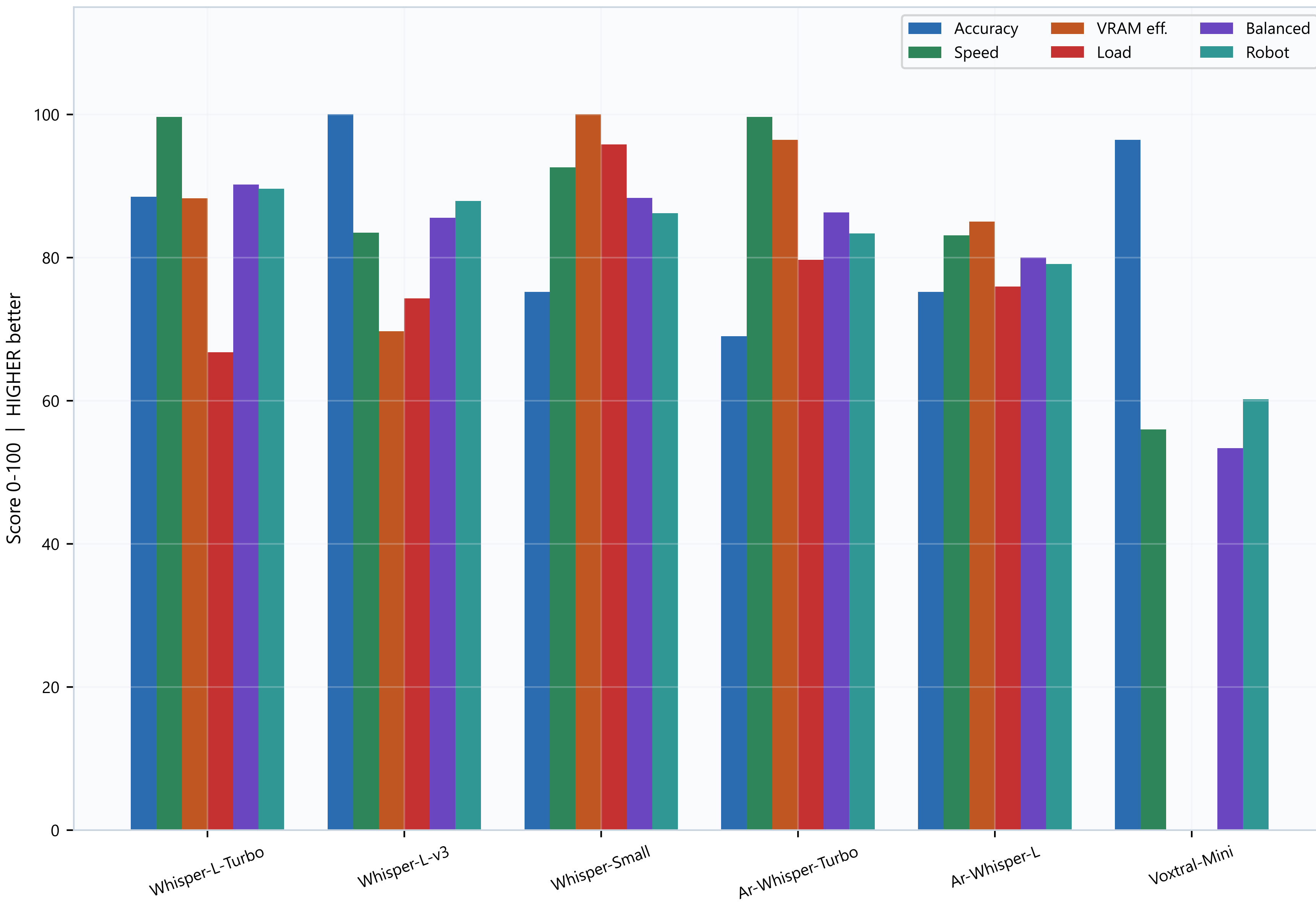
MMS-1B and Whisper Turbo variants are the speed leaders. Fast alone is not enough — pair speed charts with accuracy before selecting. A very fast but low-accuracy model will force clarification loops and hurt robot UX.

Resources (LOWER better on both charts)



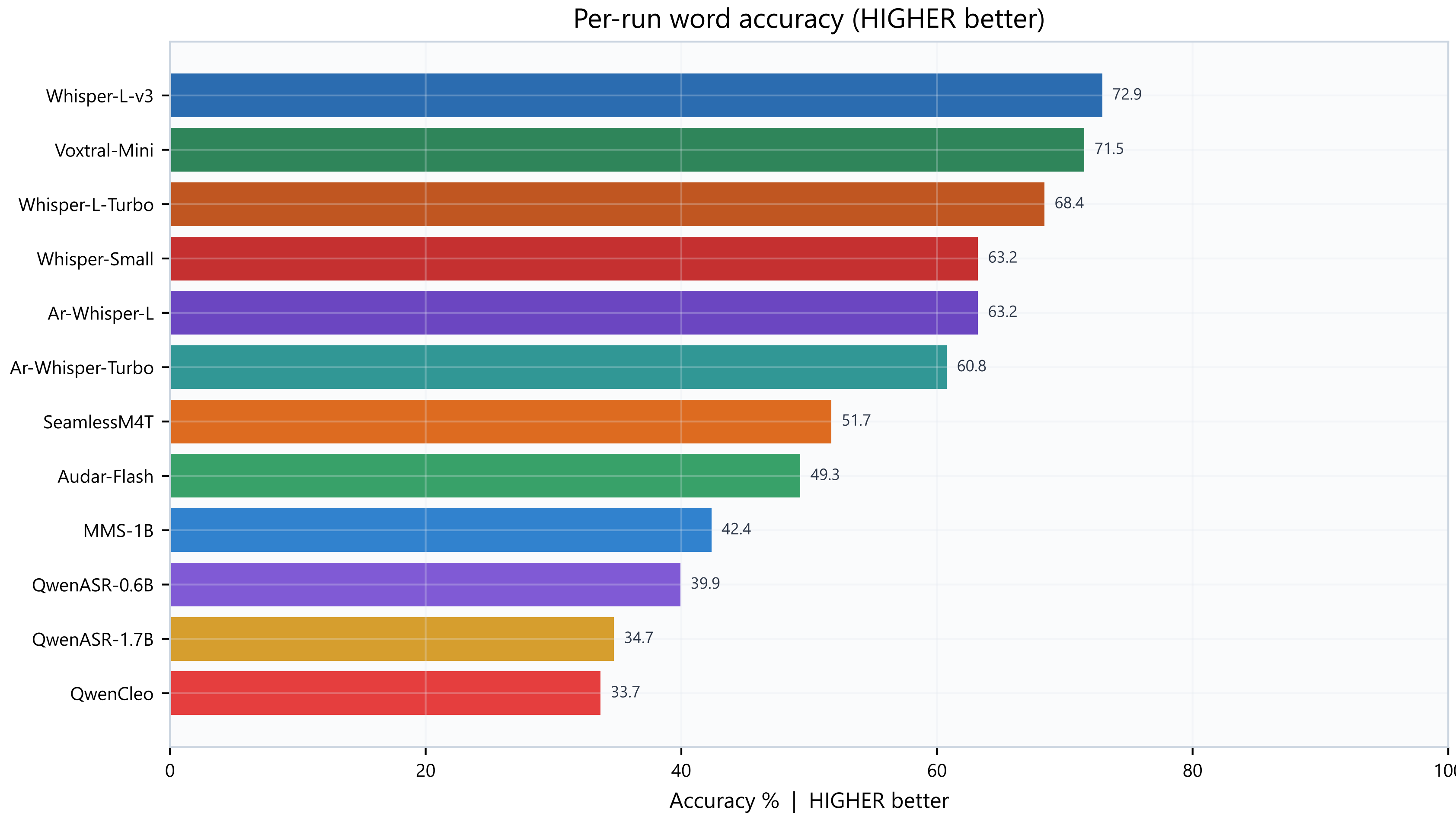
Whisper-Small is the VRAM/load champion (~1.3 GB). Whisper-Large-v3-Turbo stays under ~2.5 GB while ranking #1 on robot score — usually the best default when LLM must share the GPU. Avoid Voxtral (~11 GB) unless ASR runs alone.

Normalized score breakdown (top 6) - all HIGHER better



Use this breakdown when two models have similar robot scores. Example: Turbo wins on speed+VRAM blend; full Large-v3 wins accuracy. Pick based on whether recognition errors or latency hurt your robot more.

Accuracy ranking & selection guidance



How to choose

1) Default production pick → Whisper-Large-v3-Turbo-CT2 (best robot composite).
2) If accuracy is the hard constraint → Whisper-Large-v3-CT2.
3) If GPU is tight / multi-model → Whisper-Small-CT2 (lowest VRAM, still decent).
4) Skip Qwen ASR / QwenCleo for this Egyptian Arabic use case based on current WER.
5) Re-validate on your own microphone noise and dialect samples before freeze.